01_variables

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1 01 - PYTHON VARIABLES

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- Variable is a name assigned to a value. This name refers to a memory location.
- Python variables are also called as objects.
- Syntax: **variable = value**

variable —-> name assigned for memory location & value ——-> datatype

Rules to define a python variable name:

1. Variable names must **start with a letter or an underscore (_)**.
2. They **cannot start with a digit/number** but can end with digit/number.
3. It can contain **letters, digits, and underscores (_)** but, only if above 2 rules are met.
4. **Cannot contain spaces or special characters**.
5. They are **case-sensitive** (`python`, `Python`, and `PYTHON` are different variables).
6. It **cannot be Python keywords**. ======> Keywords are also known as reserved words.

[5]:
```
n=10 # Integer value
n
```

[5]:
```
10
```

[6]:
```
10=n # Not Valid - syntax error - Rule 2
```

```
  Cell In[6], line 1
    10=n # Not Valid - syntax error
    ^
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='
```

[8]:
```
name = 'stark' # String value
name
```

[8]:
```
'stark'
```

[9]:
```
NAME # Name Error - Rule 5, becoz variable NAME is not present (its case␣
↪sensitive - name and NAME are not the same)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[9], line 1
----> 1 NAME #

NameError: name 'NAME' is not defined
```

[10]:
```
1a = 25 # Syntax Error - Rule 2
```

```
  Cell In[10], line 1
    1a = 25
    ^
SyntaxError: invalid decimal literal
```

[11]:
```
name@ = 'tony' # Syntax Error - Rule 4
name@
```

```
  Cell In[11], line 1
    name@ = 'tony' # Name Error - Rule 4
         ^
SyntaxError: invalid syntax
```

[12]:
```
name_1 = 'tony' # Rule 3
name_1
```

[12]:
```
'tony'
```

[13]:
```
for = 45 # Syntax Error - Rule 6 (kw cannot be used as variable name)
```

```
  Cell In[13], line 1
    for = 45 # Rule 6
        ^
SyntaxError: invalid syntax
```

[15]:
```
FOR = 45
FOR
```

[15]:
```
45
```

[23]:
```
val = 53.45 # Float value
val
```

[23]:
```
53.45
```

[33]:
```
val_1 = True # Boolean value
val_1
```

[33]:
```
True
```

To display the output on the screen, `print()` function is used.

[19]:
```
print(FOR)
print(name_1)
print(name)
print(n)
```

```
45
tony
stark
10
```

******The `type()` functoin is used to identify the datatype of the variable******

[35]:
```
print(type(n))
print(type(name))
print(type(val))
print(type(val_1))
```

```
<class 'int'>
<class 'str'>
<class 'float'>
<class 'bool'>
```

To check the list of all keywords and print total number of keywords:

[37]:
```
import keyword
keyword.kwlist # Lists all the key words
```

[37]:
```
['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
```

```
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

[38]:
```
kw = len(keyword.kwlist)
print("The total number of keywords are", kw)
```

```
The total number of keywords are 35
```

[ ]:
```

```